Adrian Dar T. Serapio

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Research Interests

My research is centered on translating, monitoring, and integrating AI in clinical practice with a focus on 1) opportunistic cancer risk screening, 2) clinical context-grounded radiological interpretation, and 3) interactive clinical workflow agents, towards AI-native clinical workflows.

Education

UC Berkeley-UCSF, Berkeley & San Francisco, CA 2025–Present
Ph.D. in Bioengineering, Designated Emphasis in Computational Precision Health
Advisors: Yang Yang, Kang Wang, and Jae Ho Sohn

UC Berkeley, Berkeley, CA 2020–2024
B.S. in Electrical Engineering and Computer Sciences

Research Experience

Graduate Researcher, Yang Lab, Sohn Lab — UCSF 2025–Present

- Lead a two-stage opportunistic screening silent trial leveraging structured EHR data and routine abdominal CT scans to better screen for Pancreatic Ductal Adenocarinoma (PDAC).
- Lead the prospective clinical deployment and validation of an LLM-based summarization system with **57 radiologists** to synthesize radiologically relevant clinical context.
- Lead the development of EHRGym, a reinforcement learning environment and benchmark for training and evaluating autonomous computer-use agents in clinical workflows.

Staff Research Associate, Sohn Lab — UCSF 2024–2025

- Led a multi-reader clinical study evaluating whether LLM-generated imaging indications, synthesized from clinical notes, improve diagnostic utility compared to referring clinician-provided indications.
- Supervised undergraduate researchers in building a patient-facing vision-language agent for lung cancer screening and developing an automated evaluation metric for patient-centered radiology report communication.

Undergraduate Researcher, Sohn Lab — UCSF 2021–2024

- Led the pipeline development of an open-source, fine-tuned LLM designed to generate radiology report impressions from the radiology report findings and a multi-reader study to assess model performance against radiologists.

Undergraduate Researcher, Data-Intensive Development Lab — UC Berkeley 2020–2021

- Designed geospatial and statistical visualizations (choropleth maps, parity plots) of machine learning-based wealth estimates in Togo, which were featured in a *Nature* cover article.

Work Experience

Software Development Engineering Intern, Amazon Summer 2023

- Developed an observability pipeline for Ring Data Management to attribute infrastructure costs to internal services by aggregating time-series logs into daily and monthly cost estimates.

Selected Publications

- **Serapio, A.***, Chen, T. L.*, Tangsombatvisit, B., Fields, B. K. K., Yu, Y., Guo, Y., Kim, S. K., Miao, B. Y., Sushil, M., Hess, C. P., Majumdar, S., & Sohn, J. H. (2026). Radiologically relevant clinical history summarization with large language models: A multireader performance study. *Radiology*, 320 (2), e253238. https://doi.org/10.1148/radiol.253238
- **Serapio, A.**, Chaudhari, G., Savage, C., Lee, Y. J., Vella, M., Sridhar, S., Schroeder, J. L., Liu, J., Yala, A., & Sohn, J. H. (2024). An open-source fine-tuned large language model for radiological impression generation: A multi-reader performance study. *BMC Medical Imaging*, 24 (1), 254. https://doi.org/10.1186/s12880-024-01435-w

Technical Skills

Programming: Python, JavaScript, R, Bash
Frameworks & Tools: PyTorch, Hugging Face, NumPy, Pandas, Matplotlib, Node.js, Next.js, Git, AWS, HPC/Slurm